Unit : 1 (Vectors)

Q.1) Defination of vectors and Scalar along with example?

Ans) **Vector:-** (Cross Product)

A Vector is a quantity that has two independent properties that are magnitude and direction. The term also donates the mathematical or geometrical representation of such quantity.

Example:-

Velocity, Acceleration, Force, increase and decrease in temperature etc are examples of vector. Because these quantities have both magnitude and direction.

Scalar:- (Dot Product)

Scalar is is physical quantity that describes completely by its magnitude.

Example:- Volume, density, Speed, energy, mass etc are examples of Scalar

H-w

Q.1) Find volume of parallelepiped which has $\vec{A}, \vec{B}, \vec{C}$ as its adjacent edges where.

$\vec{A} = \hat{i} + 2\hat{j}$, $\vec{B} = 4\hat{j}$, $\vec{C} = \hat{j} + 3\hat{k}$

Let V = Volume of parallelepiped.

$V = \vec{A} \cdot (\vec{B} \times \vec{C})$

$$(\vec{B} \times \vec{C}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 4 & 0 \\ 0 & 1 & 3 \end{vmatrix}$$

$= i(12-0) - \hat{j}(0-0) - k(0-0)$

$= 12\hat{i}$

$A \cdot (\vec{B} \times \vec{C}) = (\hat{i} + 2\hat{j}) \cdot (12\hat{i})$

$= 12(\hat{i} \cdot \hat{i}) + 24(\hat{j} \cdot \hat{i})$

$= 12(1) + 24(0)$

$\vec{A} \cdot (\vec{B} \times \vec{C}) = 12$ cubic units. (Volume)

② Find Scalar triple Product:

$\vec{A} = 4\hat{i} + 2\hat{j} + \hat{k}$, $\vec{B} = \hat{i} + 5\hat{j} - 3\hat{k}$, $\vec{C} = 7\hat{i} - 6\hat{j} + 8\hat{k}$

$$(\vec{B} \times \vec{C}) = \begin{vmatrix} i & j & k \\ 1 & 5 & -3 \\ 7 & -6 & 8 \end{vmatrix}$$

$= i(40-18) - j(8+21) + k(-6-35)$

$= 22i - 29j - 44k$

$\vec{A}\cdot(\vec{B}\times\vec{C}) = (4\hat{i}+2\hat{j}+\hat{k})\cdot(22\hat{i}-29\hat{j}-44\hat{k})$

$= 88 - 58 - 41$

$= -11$ cubic unit

Q.3) Find vector triple product $\vec{A}\times(\vec{B}\times\vec{C})$, $(\vec{A}\times\vec{B})\times C$

$A = 3\hat{i} - \hat{j} + 2\hat{k}$, $B = 2\hat{i} + \hat{j} - \hat{k}$, $C = \hat{i} - 2\hat{j} + 2\hat{k}$

$\vec{A}\times(\vec{B}\times\vec{C})$

$$\vec{B}\times\vec{C} = \begin{vmatrix} i & j & k \\ 2 & 1 & -1 \\ 1 & -2 & 2 \end{vmatrix}$$

$= i(2-2) - j(4+1) + k(-4-1)$

$= -5j - 5k$

$$\vec{A}\times(\vec{B}\times\vec{C}) = \begin{vmatrix} i & j & k \\ 3 & -1 & 2 \\ 0 & -5 & -5 \end{vmatrix}$$

$= i(5+10) - j(-15-0) + k(-15-0)$

$= 15\hat{i} + 15\hat{j} - 15\hat{k}$

$(\vec{A} \times \vec{B}) \times \vec{C}$

$$\vec{A} \times \vec{B} = \begin{vmatrix} i & j & k \\ 3 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix}$$

$$= i(1-2) - j(-3-4) + k(3+2)$$

$$= -i + 7j + 5k$$

$$(\vec{A} \times \vec{B}) \times \vec{C} = \begin{vmatrix} i & j & k \\ -1 & 7 & 5 \\ 1 & -2 & 2 \end{vmatrix}$$

$$= i(14+10) - j(-2-5) + k(2-7)$$
$$= 24i + 7j - 5k$$

Characteristics Of Scalar Product:-

① If $\theta = 90^\circ$ $\quad \vec{A} \cdot \vec{B} = AB\cos(90^\circ) = 0$

$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$ (Clock wise direction) , $\hat{j} \cdot \hat{i} = \hat{k} \cdot \hat{j} = \hat{i} \cdot \hat{k} = 0$ (Anti clock wise direction)

② If $\theta = 0^\circ$ $\quad \vec{A} \cdot \vec{B} = AB\cos(0^\circ) = AB$

$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$

$\vec{A} \cdot \vec{A} = AA\cos(0^\circ) = A^2$

③ If $\theta = 180^\circ$, $\vec{A} \cdot \vec{B} = AB\cos(180^\circ) = -AB$

④ $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$

(5) $\cos\theta = \frac{\vec{A}\cdot\vec{B}}{AB} = \frac{A_xB_x + A_yB_y + A_zB_z}{AB}$

Characteristics of Vector Product:-

(1) If $\theta = 90^\circ$ $\quad \vec{A}\times\vec{B} = AB\sin 90^\circ\,\hat{n} = AB\,\hat{n}$

$i\times j = k$, $j\times k = i$, $k\times i = j$ clockwise

$j\times i = -k$, $k\times j = -i$, $i\times k = -j$ Anti clock wise.

(2) If $\theta = 0^\circ$ $\quad \vec{A}\times\vec{B} = AB\sin(0^\circ)\,\hat{n} = \vec{0}$

$\hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\cdot\hat{k} = \vec{0}$

(3) If $\theta = 180^\circ$ $\quad \vec{A}\times\vec{B} = AB\sin(180^\circ)\,\hat{n} = \vec{0}$

(4) $(\vec{A}\times\vec{B}) = (A_yB_z - A_zB_y)\hat{i} + (A_zB_x - A_xB_z)\hat{j} + (A_xB_y - A_yB_x)\hat{k}$

Note on line Integral ($\int$) :-

A line integral is an integral in which the function to be integrated is determined along a curve in the coordinate system. The function which is to be integrated may be either a scalar field or a vector field. We can integrate a scalar valued function or vector valued function along a curve.

Note on Surface Ingral $\int_s$:-

The Surface integral of a vector field F actually has a simpler explaination. If the vector field F represents the flow of a fluid, then the surface integral of F will represent the amount of fluid flowing through the surface (per unit time).

In vector calculus, the surface integral is the generalization of multiple integrals to integration over the surfaces. Sometimes, the surface integral can be thought of the double integral. For any given surface, we can integrate over surface either in the scalar field or the vector field.

Note on Volume integral ($\int_v$):-

The Volume integral is the Calculation of the volume of three-dimensional object. The Symbol for the volume integral is "$\int_v$". Volume integrals are especially important in Physics for many applications. For example, to calculate flux densities, or to Calculate mass from a Corresponding density function.

2.) What is a gradient of a Scalar function. Give its physical significance.?

Gradient of a Scalar:-

The maximum rate of increase of a Scalar function with respect to Space in a particular direction is called gradient of Scalar function.

Formula:-

$$\vec{\nabla}\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\phi$$

Physical Significance of gradient:-

The Scalar field exists in region of Space when every Point (x, y, z) of region corresponds to a Scalar function of $\phi(x,y,z)$.

The examples of Scalar field are;

① Gravitational potential energy due to mass.
② Electric potential energy due to charge.
③ Temperature.
④ Pressure.

Q) Define divergence of a vector function? Give its physical significance?

Divergence of a Vector:-

The outflow of flux from a small closed surface area per unit volume such that volume shrinks to zero is called divergence of a vector function.

Formula:-

$$\vec{\nabla}\cdot V = div\,\vec{V} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot\left(V_x\hat{i} + V_y\hat{j} + V_z\hat{k}\right)$$

$$div.\vec{V} = \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}\right)$$

Physical Significance of divergence:-

Consider a vector field $\vec{V}$ that represents a fluid velocity. The divergence of point in a fluid is a measure of the rate at which the fluid is flowing away from a point or flowing towards that point.

① Physically divergence means lines of flow diverging from a source.

② Physical divergence means lines of flow converging to a sink.

③ The divergence is zero if there is no gain or loss of lines of flow.

Q.) Define curl of a vector function or field? Give its physical significance?

Curl of a Vector field:-

Maximum value of line integral of well defined continuous and differentiable vector function around an infinitesimal region of area of surface per unit area of that surface such that this area approach to zero is called curl of a vector function or field.

Formula:-

$$\text{Curl } \vec{A} = \vec{\nabla} \times \vec{A}$$

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

Physical Significance:-

The physical Significance of a curl of a vector field is the amount of "rotation". The curl of a vector field yeilds a new vector field that describes the infinitesimal rotation at every point in the original vector field.

The direction of the vector ~~field~~ represents the direction of rotation (right hand rule) the magnitude represents how fast the sphere spins.

Gauss's Divergence Theorem:-

Statement:-

The Surface integral of normal component of a vector taken over a closed surface is equal to volume integral of divergence of that vector taken over the volume inclosed by that Surface.

Surface integral of normal component of $\vec{A}$. = Volume integral of divergence of $\vec{A}$

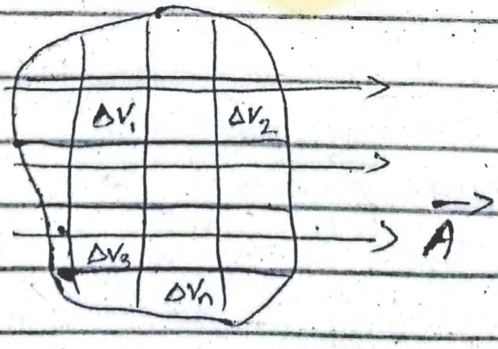
Mathematically:-

$$\int_S \vec{A}\cdot\vec{da} = \int_V (\vec{\nabla}\cdot\vec{A})\,dV$$

Proof:-

Consider a Surface "S" having Volume (V) Placed in a vector $\vec{A}$ field. Divide the volume into Small element $(\Delta V_1, \Delta V_2 ---$ The Volume element is enclosed by Surface element $(\Delta S_1, \Delta S_2, \Delta S_3 ----)$.

Diagram:-



The out flow of flux comming from ith Surface element ΔS_i having area da_i is given as;

$$\Delta\phi_i = \int_{\Delta S_i} \vec{A} \cdot \vec{da_i} \quad \text{——— (1)}$$

The Net flux from whole surface (S) is;

$$\text{Net flux} = \int_S \vec{A} \cdot \vec{da} \quad \text{——— (2)}$$

The net flux ~~thro~~ due to all surface element can also be written as;

$$\text{Net flux} = \Delta\phi_1 + \Delta\phi_2 + \Delta\phi_3 + \cdots + \Delta\phi_n$$

OR

$$\text{Net flux} = \sum_{i=1}^{N} \Delta\phi_i \quad \text{——— (3)}$$

Put Eq (1) in Eq (3)

$$\text{Net flux} = \sum_{i=1}^{N} \int_{\Delta S_i} \vec{A} \cdot \vec{da_i}$$

xing and $\div$ing R.H.S by ΔV_i

$$= \left[\sum_{i=1}^{N} \int_{\Delta S_i} \frac{\vec{A} \cdot \vec{da_i}}{\Delta V_i} \right] \Delta V_i \quad \text{——— (4)}$$

In limiting case $\sum_{i=1}^{N} \longrightarrow \int$ and $\Delta V_i \longrightarrow dV$

while $\lim_{\Delta V \to 0} \left(\int \frac{\vec{A} \cdot \vec{da_i}}{\Delta V_i} = \vec{\nabla} \cdot \vec{A} \right)$

hence Eq (4) becomes.

$$\text{Net flux} = \int_v (\vec{\nabla}\cdot\vec{A})\,dV \quad \text{——— (5)}$$

Comparing Eq (2) & (5)

$$\int_s (\vec{A}\cdot\vec{da}) = \int_v (\vec{\nabla}\cdot\vec{A})\,dV$$

Hence Proved.

Stoke's Theorem:-

This theorem enables us in some cases to go from Surface integral to a line integral.

Statement:-

The line integral of a vector function around any closed curl is equal to the Surface integral of curl of that function taken over the Surface enclosed by the curve at this mathematical boundary.

line integral of vector function = Surface integral of curl or that function.

Mathematically:-

$$\oint \vec{A}\cdot\vec{dr} = \int_S (\vec{\nabla}\cdot\vec{A})\,da$$

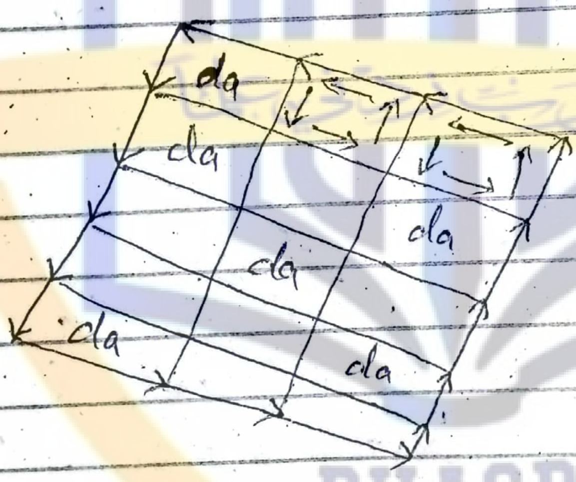
Proof:-

Consider a closed Surface (S) with its boundry as a closed curve. The curl is taken in anticlockwise direction, so the line integral around the curve is represented as;
$$\oint \vec{A} \cdot \vec{dr}.$$

Where $\vec{dr}$ is the element of path length. We devide the surface into small elements "da" so that;

$$\vec{da} = \hat{n}\, da$$

Diagram:-



The line integral along the common sides of adjacent element will cancel each other and we are only left with the contribution from the sides lying on boundary.

Since curl of a vector is a line integral per unit area.

Hence curl of $\vec{A}$ ~~(vec)~~ taken over this Surface is;

$$\text{Curl } \vec{A} = \frac{\oint \vec{A} \cdot \vec{dr}}{\int \hat{n} da}$$

$$\text{curl } \vec{A} \int \hat{n} da = \oint \vec{A} \cdot \vec{dr}$$

which is the Stoke's theorem.

Parallel Axis Theorem:-

Statement:-

Moment of inertia of a rigid body about any arbitrary axis is equal to moment of inertia about an axis passing through the centre of mass plus total mass times square of the distance between them.

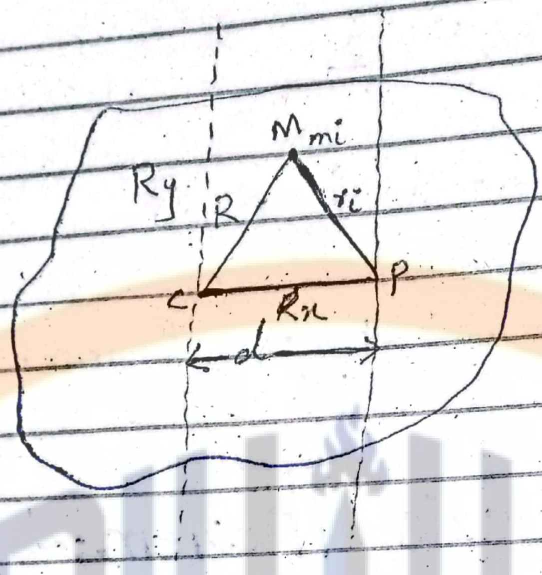
Mathematically:-

$$I = I_{cm} + Md^2$$

Proof:-

Consider a rigid body whoose moment of inertia is to be found. where "C" is the centre of mass and "m_i" is mass particle of rigid body.

Diagram:-



Moment of inertia about centre of mass is;

$$I_{cm} = \sum m_i R^2 \quad \text{—— (1)}$$

where;

$$R = \frac{\sum m_i r_i^2}{M}$$

$$R_x = \frac{\sum m_i x_i^2}{M}, \quad R_y = \frac{\sum m_i y_i^2}{M}$$

Rotation about point 'P' gives moment of inertia.

$$I = \sum m_i r_i^2 \quad \text{—— (2)}$$

Using laws of cosine, Putting in eq (2)

$$r_i^2 = R^2 + d^2 - 2Rd\cos\theta$$

$$I = \sum m_i (R^2 + d^2 - 2Rd\cos\theta)$$

$$I = \sum m_i R^2 + \sum m_i d^2 - \sum m_i \, 2Rd\cos\theta$$

$$I = R^2 \sum m_i + d^2 \sum m_i - 2d \sum m_i R\cos x$$

$$I = R^2 M + d^2 M - 2d \sum m_i R_x$$

If we take centre of mass on origin, where x_x and x_y becomes zero.

Hence

$$I = R^2 M + d^2 M$$

$$I = \sum m_i R^2 + Md^2$$

Using eq ① we have.

$$I = I_{cm} + Md^2$$

Hence parallel axis theorem is proved.

Important Theorems

① If $\vec{A} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$, find divergence of A at point $1, -1, 1$

$$\vec{A} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$$

$$\vec{\nabla}\cdot\vec{A} = ?$$

$$\vec{\nabla}\cdot\vec{A} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\cdot\left(xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}\right)$$

$$= \frac{\partial}{\partial x}(xz^3) + \frac{\partial}{\partial y}(-2x^2yz) + \frac{\partial}{\partial z}(2yz^4)$$

$$\vec{\nabla}\cdot\vec{A} = z^3 - 2x^2z + 8yz^3$$

value of $\vec{\nabla}\cdot\vec{A}$ at $\overset{x}{(1}, \overset{y}{-1}, \overset{z}{1)}$

$$\vec{\nabla}\cdot\vec{A} = (1)^3 - 2(1)^2(1) + 8(-1)(1)^3$$

$$= 1 - 2 - 8$$

$$\boxed{\vec{\nabla}\cdot\vec{A} = -9}$$

② Prove that gradient $S = \nabla S$

we know that

$$\nabla S = \frac{\partial s}{\partial x}\hat{i} + \frac{\partial s}{\partial y}\hat{j} + \frac{\partial s}{\partial z}\hat{k}$$

Now,

$$\vec{\nabla S}\cdot\vec{dr} = \left(\frac{\partial s}{\partial x}\hat{i} + \frac{\partial s}{\partial y}\hat{j} + \frac{\partial s}{\partial z}\hat{k}\right)\cdot\left(dx\hat{i} + dy\hat{j} + d_z\hat{k}\right)$$

$$\nabla S \cdot dr = \frac{\partial s}{\partial x} dx + \frac{\partial s}{\partial y} dy + \frac{\partial s}{\partial z} dz \quad \text{——— (1)}$$

By Defination.

$$\text{grad } S = \frac{ds}{dn} \hat{n}$$

$$\text{grad } S \cdot \vec{dr} = \frac{ds}{dn} \hat{n} \cdot \vec{dr}$$

$$= \frac{ds}{dn} |\hat{n}| |\vec{dr}| \cos\theta = \frac{ds}{dn} dr \cos\theta$$

$$\frac{dn}{dr} = \cos\theta, \quad dn = dr \cos\theta.$$

So,

$$\text{grad } S \cdot \vec{dr} = \frac{ds}{dn} dn = ds.$$

In rectangular coordinates

$$ds = \frac{\partial s}{\partial x} dx + \frac{\partial s}{\partial y} dy + \frac{\partial s}{\partial z} dz$$

Therefore.

$$\text{grad } S \cdot \vec{dr} = \frac{\partial s}{\partial x} dx + \frac{\partial s}{\partial y} dy + \frac{\partial s}{\partial z} dz \quad \text{——— (2)}$$

Comparing Eq (1) & (2), we get

$$\text{grad } S \cdot \vec{dr} = \nabla S \cdot \vec{dr}$$

$$\boxed{\text{grad } S = \nabla S}$$

(3) Prove that grad $\vec{r} = \frac{\vec{r}}{r}$

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\vec{r}\cdot\vec{r} = (x\hat{i}+y\hat{j}+z\hat{k})\cdot(x\hat{i}+y\hat{j}+z\hat{k})$

$r^2 = x^2+y^2+z^2$

Diff w.r.t x

$2r\frac{dr}{dx} = 2x \Rightarrow \boxed{\frac{\partial r}{\partial x} = \frac{x}{r}}$

Similarly, $\boxed{\frac{\partial r}{\partial y} = \frac{y}{r}}$ & $\boxed{\frac{\partial r}{\partial z} = \frac{z}{r}}$

Now,

grad $r = \nabla r = \frac{\partial r}{\partial x}\hat{i} + \frac{\partial r}{\partial y}\hat{j} + \frac{\partial r}{\partial z}\hat{k}$

$= \frac{x}{r}\hat{i} + \frac{y}{r}\hat{j} + \frac{z}{r}\hat{k}$

$= \frac{x\hat{i}+y\hat{j}+z\hat{k}}{r}$

$\boxed{\text{grad } \vec{r} = \frac{\vec{r}}{r}} \Rightarrow \hat{r}$

(4) Prove that grad $\frac{1}{r} = \frac{-\vec{r}}{r^3}$

grad $\frac{1}{r} = \frac{-\vec{r}}{r^3}$

$\frac{\partial r}{\partial x} = \frac{x}{r}$, $\frac{\partial r}{\partial y} = \frac{y}{r}$, $\frac{\partial r}{\partial z} = \frac{z}{r}$

Now,

$$\text{grad}\left(\frac{1}{r}\right) = \frac{\partial}{\partial x}\left(\frac{1}{r}\right)\hat{i} + \frac{\partial}{\partial y}\left(\frac{1}{r}\right)\hat{j} + \frac{\partial}{\partial z}\left(\frac{1}{r}\right)\hat{k}$$

$$= -\frac{1}{r^2}\frac{\partial r}{\partial x}\hat{i} - \frac{1}{r^2}\frac{\partial r}{\partial y}\hat{j} - \frac{1}{r^2}\frac{\partial r}{\partial z}\hat{k}$$

$$= -\frac{1}{r^2}\left(\frac{\partial r}{\partial x}\hat{i} + \frac{\partial r}{\partial y}\hat{j} + \frac{\partial r}{\partial z}\hat{k}\right)$$

$$= \frac{-1}{r^2} \cdot \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{r}$$

$$= \frac{-1}{r^2} \cdot \frac{\vec{r}}{r}$$

$$\boxed{\text{grad}\,\frac{1}{r} = \frac{-\vec{r}}{r^3}}$$

(5) Prove that div grad $(\phi) = \nabla^2\phi$

$$(\text{div grad}\,\phi) = \left[\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right] \cdot \left(\frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k}\right)$$

$$= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

$$= \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right]\phi$$

$$\text{div}(\text{grad}\,\phi) = \nabla^2\phi$$

(6) Show that $\text{curl}(\text{grad}\ \phi) = 0$

$$\text{Curl}(\text{grad}\ \phi) = \nabla \times \nabla\phi$$

$$\text{curl}(\text{grad}\ \phi) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial\phi}{\partial x} & \frac{\partial\phi}{\partial y} & \frac{\partial\phi}{\partial z} \end{vmatrix}$$

$$= \left(\frac{\partial^2\phi}{\partial y\partial z} - \frac{\partial^2\phi}{\partial y\partial z}\right)\hat{i} + \left(\frac{\partial^2\phi}{\partial x\partial z} - \frac{\partial^2\phi}{\partial x\partial z}\right)\hat{j} + \left(\frac{\partial^2\phi}{\partial x\partial y} - \frac{\partial^2\phi}{\partial x\partial y}\right)\hat{k}$$

$$\text{Curl}(\text{grad}\phi) = 0 \qquad \boxed{\text{Curl}(\text{grad}\ \phi) = 0}$$ Proved

(7) Prove that: $\text{div}(\text{curl}\ \vec{F}) = 0$

$$\text{div}(\text{Curl}\ \vec{F}) = \nabla \cdot \nabla \times \vec{F}$$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$= \left[\frac{\partial}{\partial y}\left\{F_z - \frac{\partial}{\partial z}F_y\right\}\hat{i} + \left\{\frac{\partial}{\partial x}F_z - \frac{\partial}{\partial z}F_x\right\}\hat{j} + \left\{\frac{\partial}{\partial x}F_y - \frac{\partial}{\partial y}F_x\right\}\right]$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) + \frac{\partial}{\partial y}\left(\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z}\right) + \frac{\partial}{\partial z}\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)$$

$$= \cancel{\frac{\partial^2 F_z}{\partial x \partial y}} - \cancel{\frac{\partial^2 F_y}{\partial x \partial z}} + \cancel{\frac{\partial^2 F_z}{\partial y \partial x}} - \cancel{\frac{\partial^2 F_x}{\partial y \partial z}} + \cancel{\frac{\partial^2 F_y}{\partial z \partial x}} - \frac{\partial^2 F_x}{\partial z \partial y}$$

$div(curl\, \vec{F}) = 0$

(8) If $\phi = 2x^3y^2z^4$ Find $div(grad\ \phi)$

$$div(grad\ \phi) = \nabla \cdot \nabla\phi$$

$$\nabla \cdot \nabla\phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

$$= \frac{\partial^2}{\partial x^2}(2x^3y^2z^4) + \frac{\partial^2}{\partial x^2}(2x^3y^2z^4) + \frac{\partial^2}{\partial x^2}(2x^3y^2z^4)$$

$$\nabla \cdot \nabla\phi = 12xy^2z^4 + 4x^3yz^4 + 24x^3y^2z^2$$

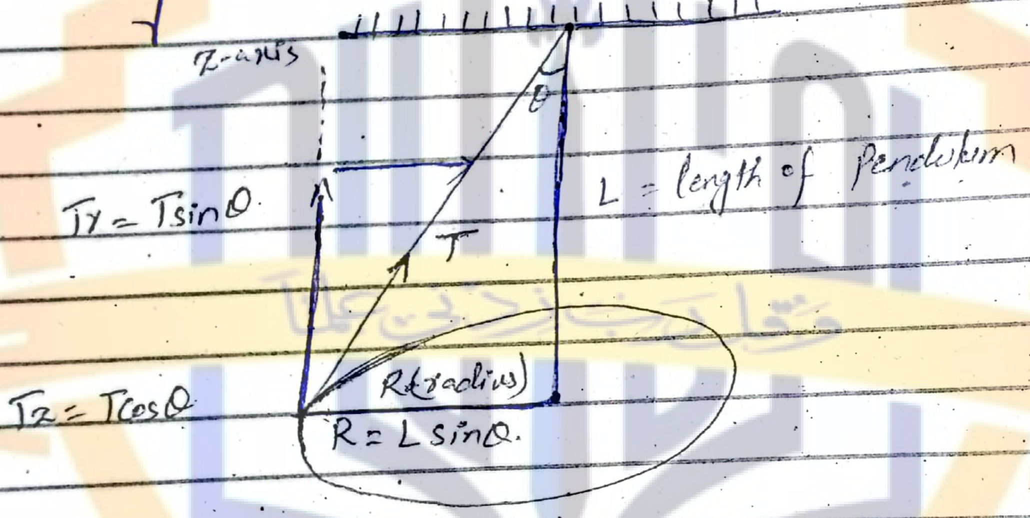
$$\nabla^2\phi = 12xy^2z^4 + 4x^3z^4 + 24x^3y^2z^2$$

Conical Pendulum

Statement:~

The bob having mass "m" fixed at the end of a String of length(L) when rotated in horizontal circle having radius "r" with constant speed "v" such that String describes a cone called conical pendulum.

Diagram:~



Explaination:~

Let us find the time period of pendulum. If the string makes an angle "θ" with the vertical line, then the radius of circular path is;

$$R = L \sin\theta .$$

There are two forces acting on the body, one is the tension acting along the string and the other is weight acting vertically downward. Resolving tension into components are;

$$T_x = -T\sin\theta \quad \text{---} (1)$$

Negative sign indicates that force is directed towards the centre of circle providing it the centripital force and "z" component is.

$$T_z = T\cos\theta \quad \text{---} (2)$$

Since "T_z" is balanced with weight (mg) hence we may write;

$$T_z = mg \quad \text{---} (3)$$

Comparing (2) & (3), we have.

$$T\cos\theta = mg \quad \text{---} (4)$$

The centripital force is given by;

$$F_x = T_x = \frac{-mv^2}{R} \quad \text{---} (5)$$

From eq (1)

$$T_x = -T\sin\theta$$

Comparing (1) & (5)

$$\cancel{-}T\sin\theta = \frac{\cancel{-}mv^2}{R}$$

$$T\sin\theta = \frac{mv^2}{R} \quad (6)$$

$\div$ing eq (6) by eq (4)

$$\frac{T\sin\theta}{T\cos\theta} = \frac{mv^2}{mgR}$$

$$\frac{\sin\theta}{\cos\theta} = \frac{v^2}{gR}$$

$$\tan\theta = \frac{v^2}{Rg}$$

$$v = \sqrt{\tan\theta\, Rg} \quad (7)$$

Period of Revolution:-

It is the time in which a body completes one revolution.

$$\text{time period} = \frac{\text{distance travelled in 1 rev}}{\text{speed of the body}}.$$

$$T = \frac{2\pi R}{v} \quad (8)$$

Putting the value of v from eq (7)

$$T = \frac{2\pi R}{\sqrt{Rg\tan\theta}}$$

Since, $R = \sqrt{R}\cdot\sqrt{R}$

$$T = \frac{2\pi}{\sqrt{R}}\,\frac{\sqrt{R}\cdot\sqrt{R}}{\sqrt{g\tan\theta}}$$

$$T = \frac{2\pi\sqrt{R}}{\sqrt{g\tan\theta}}$$

Since

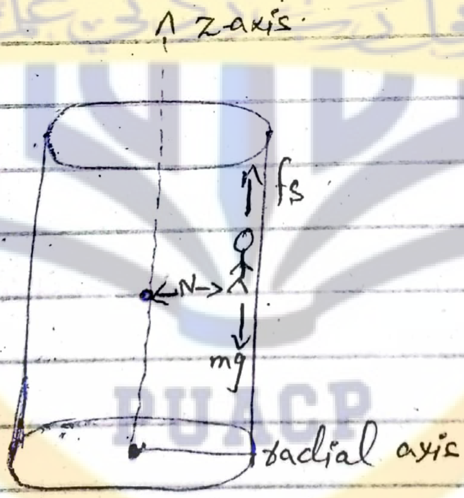
$$R = L\sin\theta$$

$$T = 2\pi\sqrt{\frac{L\sin\theta}{g\sin\theta}} \cdot \sqrt{\cos\theta}$$

$$T = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

This is the time period of conical pendulum.

The Rotor:~



A rotor is a hollow cylindrical room that can be set into rotation about the central vertical axis of the cylinder.

Suppose we consider a person standing up against a wall, the rotor is rotated with a speed "V" and floor below the person is opened down, the person does not fall but remain pinned up against the wall of rotor.

Now we calculate the maximum rotational speed to prevent the person from falling.

Forces acting on a person are;

① Force of static friction (f_s) acting vertically upward.

② weight of person (mg) acting vertically downward.

3) Normal reaction of wall (N), acting normal to the wall i.e by the wall on the person.

Now consider the vertical direction as "z" axis and horizontal direction as radial axis. Thus the vertical component of the force is;

$$f_s = mg$$

Since, $f_s = \mu_s N$

$$\mu_s N = mg$$

$$N = \frac{mg}{\mu_s} \quad \text{—— ①}$$

This is the normal force (N) directed from wall towards men.

It provides the necessary centripital force on men moving in a circular path of radius (R) and is given as:-

$$N = \frac{mv^2}{R} \quad \text{———} (2)$$

Comparing (1) & (2)

$$\frac{mg}{\mu_s} = \frac{mv^2}{R}$$

$$v = \sqrt{\frac{gR}{\mu_s}} \quad \text{———} (3)$$

As $v = R\omega$, so putting values in eq (3) we have;

$$R\omega = \sqrt{\frac{gR}{\mu_s}}$$

$$\cancel{\sqrt{R}} \cdot \sqrt{R}\,\omega = \frac{\sqrt{g}\,\cancel{\sqrt{R}}}{\sqrt{\mu_s}}$$

$$\omega = \sqrt{\frac{g}{\mu_s R}}$$

This is the minimum speed of rotation which prevents the men from falling downward.

Banked Curve/Track :-

Statement:-

The curve in which a vehicle banks or inclines towards inside of the curve for safe turn is called Banked curve.

The angle at which the vehicle is inclined about its longitudinal axis with respect to its path is called bank/banking angle.

Explaination:-

A centripital force is needed to keep a car in its curved track when it takes a turn. This centripital force is provided by the friction between tires and the road. The car would slip if the force of friction is not sufficient, this problem is solved by banking the curved road, i.e the outer edge of the road is raised.

Consider a car of mass "m" moving with a speed 'v' on a level road around a curved path having radius of curvature (R). The necessary centripital force for car to move in a ~~(circular)~~ circle, provided by frictional force is given by:

$$f = \frac{mv^2}{R}$$

$$\mu N = \frac{mv^2}{R}$$

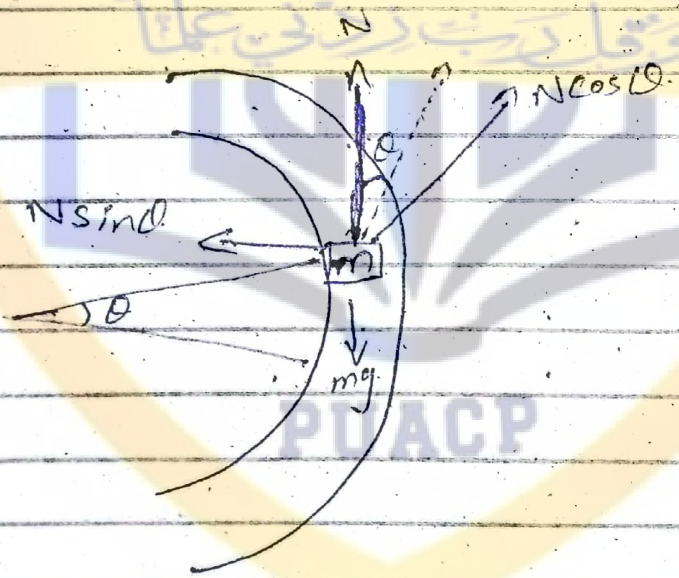
$$\mu \cancel{m} g = \frac{\cancel{m} v^2}{R}$$

$$\mu g = \frac{v^2}{R}$$

$$v = \sqrt{\mu g R}$$

This is the cornering speed. This must be large enough to provide centripital force.

Frictionless Banked Curve:-



Consider a car of mass (m) moving on a track with banking angle (θ) with respect to inner edge for save turn since force of friction is absent.

The forces acted on the car on banked curved are;
① weight (mg) acting vertically downward.
② Normal force (N) acting vertically upward on the car.
The banked inclined edge provides additional force to keep the car in its path and prevent from pushing in or out of the circular path. This force is $N\sin\theta$ & it provides neccessary centripital force. Therefore

$$N\sin\theta = \frac{mv^2}{R} \quad \text{——— ①}$$

The component $N\cos\theta$ is balanced by weight of the car, hence,

$$N\cos\theta = mg \quad \text{——— ②}$$

÷ing ① & ②

$$\frac{\cancel{N}\sin\theta}{\cancel{N}\cos\theta} = \frac{\cancel{m}v^2}{\cancel{m}gR}$$

$$\tan\theta = \frac{v^2}{gR}$$

$$\boxed{v = gR\tan\theta} \quad \text{——— ③}$$

It is called rated speed of a turn.
Eq ③ shows that this speed is independent of mass of car and is same for all massive objects.
It depends on the banking angle and radius of curvature.